Academic Year 2026–2027 · Geomatics & Spatial Modelling (GMS) · Master 2

FME Workflow & Spectral Indices — Project

Building Enrichment Workflow: BD TOPO® BATIMENT + LiDAR Heights + LST, NDVI & other related Spectral Statistics from Landsat 8/9 OLI/TIRS

Cours Assignment · Group work based on groups created during the last session (group change is not allowed) · Weights on final grade & real application of the processed data in energy performance modelling of the 13e arrondissement de Marseille (13013).

Summary. You will design and run an FME workspace that ingests the BD TOPO® BATIMENT layer, enriches each building footprint with its **height derived from LiDAR**, and with the **mean, maximum and minimum** values of **LST**, **NDVI**, **NDBI**, **NDMI**, **NDWI**, **BSI**, **emissivity** and **Albedo** computed over the footprint. **Building height extraction using LiDAR should be included in the FME workflow**, while spectral indices can be computed using a Python script and then added to the FME output as an additional data attribute. Export the clipped result only for the 13e arrondissement de Marseille as a single ESRI shapefile respecting the BD TOPO® BATIMENT attribute schema.

1. Learning Objectives

On completion of this project you will be able to:

- Build a reproducible spatial data integration workflow in FME (Safe Software).
- Manipulate the BD TOPO® BATIMENT schema and its coded attribute domains.
- Extract building heights from a LiDAR point cloud using zonal statistics on DSM/DTM.
- Compute LST from a Landsat thermal band and NDVI + other spectral indices as previously mentioned, from multispectral reflectance.
- Validate, document and deliver a compliant shapefile product.

2. Context & Data Sources

The **BD TOPO®** is the topographic reference database of the IGN (Institut national de l'information géographique et forestière, France). Its [BATIMENT] layer describes building footprints with a rich attribute model. For this assignment you will enrich that layer with LiDAR and raster-derived indicators.

The **LiDAR**, IGN Lidar HD is a classified point cloud, used to derive a Digital Surface Model (DSM) and Digital Terrain Model (DTM); building height = Z_DSM − Z_DTM over the footprint.

The **LST** and all other spectral indices must be derived and computed from Landsat 8/9 OLI/TIRS surface reflectance products. These indices must be added to the FME output shapefile and should contain four predefined dates, meaning that you need to calculate these indices for all of these dates (**2013_07_23**, **2013_12_30**, summer of **2026**, **2025_11_29**).

The land cover and land use data from OpenStreetMap (OSM) should be used to derive building types and land use-related missing data in BD TOPO via workflow integration.

Within the FME pipeline, the entire workflow should proceed with respect to the boundary of the 13e arrondissement de Marseille, only, and there should be no external manipulation outside of FME, except for spectral indices. One final executable FME workflow is required to generate the final Enriched_Building.shp file.

All datasets must share a common projected CRS (recommend EPSG:2154 - Lambert 93).

3. Required Output: Shapefile Attribute Schema

The final shapefile must contain exactly the attributes below, in this order, with the specified type. Rename the original fields as requested in the table (marked as ; →), and create any required fields. Note that, for fields containing missing values, these should be calculated and filled in based on assumptions derived from official statistics (e.g. usage, NB_LOGTS, NB_ETAGES, population). Ensure that there are no missing values wherever possible, and if this is not possible, explain and document the reason why.

Field	DataType	Description
BD TOPO® BATIMENT attributes + generated fields		
ID	Integer	Unique feature identifier (first field)
NATURE → (nature)	Text(50)	Nature of the building (coded)
USAGE1 → (building_type_1)	Text(20)	Primary usage (coded)
USAGE2 → (building_type_2)	Text(20)	Secondary usage (coded)
DATE_APP → (date)	Date	Appearance date
area_m2	Double	Area of builidings
perimeter_m	Double	Perimeter of buildings
building volume_m3	Float	Building Volume (m^3)
longside_angle	varchar	Building long side angle
wwr	Float	Window-wall-ratio of 20 %
Total_floor_area_m2	Double	Building total fllor area_m2
wall_area_m2	Double	Building external envelope area_m2
Building_Compactness_Index	---	Cube-referenced relative compactness (BCI), use this reference paper to calculate it. DOI: 10.1016/j.enconman.2015.06.011
Occupancy	Double	Nb of people living in a building based on assumptions

NB_LOGTS → (nb_appart)	Integer	Number of dwellings
NB_ETAGES → (nb_floor)	Integer	Number of floors
MAT_MURS→(wall_material)	Text(20)	Wall material (coded)
MAT_TOITS→(roof_material)	Text(20)	Roof material (coded)
Z_MIN_SOL →(zmin_ground)	Real	Minimum ground elevation over the footprint (m)
Z_MAX_SOL→(zmax_ground)	Real	Maximum ground elevation over the footprint (m)
Z_MIN_TOIT → (zmin_roof)	Real	Minimum roof elevation over the footprint (m)
Z_MAX_TOIT → (zmax_roof)	Real	Maximum roof elevation over the footprint (m)
HAUTEUR → (height_m)	Real	Building height from BD TOPO® (m)
LiDAR height derived field		
Height_m_LIDAR	Real	Building height = max(DSM − DTM) over the footprint (m)
Thermal & spectral statistics per footprint (mean / max / min) in separate fields		
LST (MEAN / MAX / MIN)	Real	Land Surface Temperature (°C)
NDVI (MEAN / MAX / MIN)	Real	Normalized Difference Vegetation Index
NDBI (MEAN / MAX / MIN)	Real	Normalized Difference Built-up Index
NDMI (MEAN / MAX / MIN)	Real	Normalized Difference Moisture Index
NDWI (MEAN / MAX / MIN)	Real	Normalized Difference Water Index
BSI (MEAN / MAX / MIN)	Real	Bare Soil Index
EMIS (MEAN / MAX / MIN)	Real	Surface emissivity (NDVI-based)
ALB (MEAN / MAX / MIN)	Real	Surface albedo (0–1)

4. Deliverables

1. **Enriched_Building.shp** - the final shapefile (with .shx/.dbf/.prj/.cpg).
2. **workflow.fmw** - the FME workspace (runnable, commented).
3. **Python code** - the clean and executable python scripts with comments and formulas.
4. **report.pdf** (2–5 pages) - methodology, transformers used, LST/NDVI … formulas, results, screenshots of the clean FME workspace and the output attribute table.
5. **README.txt** - data paths, FME workspace information, input/output details.
6. A 5-minute presentation (slides) - summarising the workflow and results.

Submit all files as a single **.zip** archive named `Data_Manipulation_Project_GroupNumber.zip` via email or a downloadable link.

Wish you all the best and every success in your studies!

FME Project Report - Enrichment of the TOPO database on the 13th arrondissement of Marseille

1. Project Objective

The objective of this project is to design and execute an automated spatial data processing workflow using FME (Safe Software). This workflow aims to enrich the "BUILDING" layer of the IGN TOPO® database on the strict footprint of the 13th arrondissement of Marseille (13013). This enrichment is multidimensional: it integrates elevation data from the IGN's HD LiDAR to calculate the height of buildings, participatory data from OpenStreetMap (OSM) to complete the usage information, as well as spectral indices from Landsat 8/9 satellite imagery to characterize the radiative and thermal environment of each structure. The final result is exported in the form of a single Shapefile (Enriched_Building.shp) respecting a precise attribute scheme, intended for energy performance modeling. The data set is projected into the Lambert 93 reference coordinate system (EPSG:2154).

2. Methodology and Spatial Processing Chain

The workflow is structured around two distinct phases: external preprocessing in Python and spatial integration in FME.

2.1. External processing (Python)

Spectral indices requiring complex matrix manipulations on the Landsat 8/9 bands were computed via a Python script upstream of FME. Eight indices were generated: LST (Land Surface Temperature), NDVI, NDBI, NDMI, NDWI, BSI, Emissivity and Albedo. These calculations were applied on four separate dates required by the specifications: 23/07/2013, 30/12/2013, 29/11/2025 and 19/08/2026.

2.2. FME Integration Flow

In FME, the process begins with the import of the BD TOPO® layer and its cutting (Clipper/SpatialFilter) on the administrative boundary of the 13th arrondissement. For altimetry, a zonal analysis is performed on the classified LiDAR point cloud, converted into numerical models. The height of each building is extracted directly in FME by calculating the difference between the maximum roof elevation and the minimum ground elevation on the building footprint (Z_DSM - Z_DTM). Then, a spatial join is performed with the rasters generated by the Python script. FME extracts for each building footprint the spectral statistical values (Average, Maximum and Minimum) for the 8 indices and the 4 dates. Finally, the OpenStreetMap (OSM) data are cross-referenced with the geometric footprints to deduce the types of buildings and the land cover where the BD TOPO is incomplete.

3. FME transformers used

To carry out this processing, the FME workflow relies on several categories of transformers:

- **Readers and Readers:** Set to read TOPO comics, HD LiDAR tiles, OSM files and spectral rasters with on-the-fly reprojection in EPSG:2154.
- **Filters and geometry:** SpatialFilter and Clipper to strictly restrict processing to the study area. AreaCalculator and VolumeCalculator to extract basic polygon metrics.

- **Raster and LiDAR Analysis:** ZonalStatisticsCalculator (or a combination of RasterBandMinMaxExtractor / PointOnAreaOverlayer depending on the conversion) to extract LiDAR heights and statistics from spectral rasters.
- **Schema manipulation:** AttributeManager, AttributeCreator, and AttributeRenamer to clean up the attribute table structure, rename fields as required, and estimate missing values. NullAttributeMapper to standardize empty values.

4. Mathematical Formulations and Clues

Normalized Vegetation Index (NDVI)

NDVI = (PIR - Red) / (PIR + Red)

Normalized Difference Index of Buildings (NDBI)

NDBI = (SWIR1 - PIR) / (SWIR1 + PIR)

Normalized Humidity Index (NDMI)

NDMI = (PIR - SWIR1) / (PIR + SWIR1)

Normalized Differential Water Index (NDWI)

NDWI = (Green - PIR) / (Green + PIR)

Bare Soil Index (BSI)

BSI = ((SWIR1 + Red) - (PIR + Blue)) / ((SWIR1 + Red) + (PIR + Blue))

Surface Emissivity (ε)

$\varepsilon = 0.004 * Pv + 0.986$ (where $Pv = [(NDVI - NDVI_{min}) / (NDVI_{max} - NDVI_{min})]^2$)

Surface Albedo

Albedo = (0.356 * B2) + (0.130 * B3) + (0.373 * B4) + (0.085 * B5) + (0.072 * B6) - 0.0018

Land Surface Temperature (LST)

The temperature is extracted from the thermal band (Band 10) converted to brightness temperature (Tb) and then corrected with emissivity (ε).

$LST = Tb / [1 + (\lambda * Tb / \rho) * \ln(\varepsilon)]$ *(Where λ is the wavelength of the thermal band and ρ is a physical constant based on Planck's constant).*

Building Compactness Index (BCI)

Morphological indicator calculated directly in FME to assess energy efficiency based on the geometric shape of the footprint.

$BCI = (2 * \sqrt{(\pi * Area)}) / Perimeter$

5. Attribute Schema Management and Missing Values

The final deliverable is exported as an ESRI Shapefile (Enriched_Building.shp). Because the Shapefile format is based on the DBF standard, a strict constraint of up to 10 characters has been

applied to the attribute names. For example, the initial field "USAGE1" has been renamed to "bldg_type1" (or "build_typ1").

Particular attention has been paid to the handling of missing values through the use of the AttributeManager:

- **NB_ETAGES (Number of Floors):** When absent, it is estimated by dividing the height calculated via the LiDAR by a standard floor height of 3 meters (Hauteur_LiDAR / 3).
- **NB_LOGTS (Number of dwellings):** For residential buildings without this data, an estimate is made by dividing the estimated total floor area (Floor area * NB_ETAGES) by a theoretical average surface area of an apartment (e.g. 70 m^2).
- **Occupancy:** The theoretical number of inhabitants was calculated based on the volume of the building and the main use deduced from OSM data (residential, tertiary, etc.).

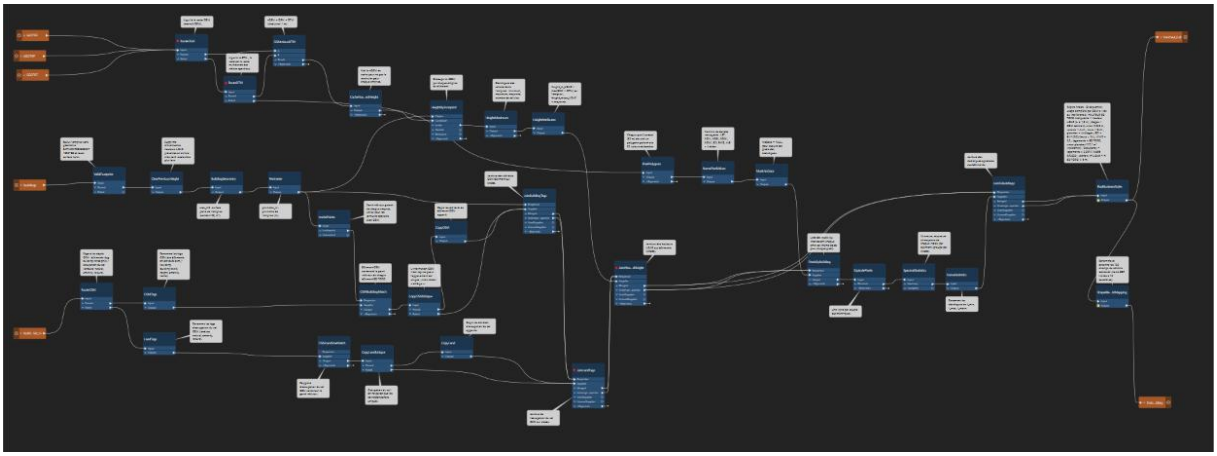


Figure 1: Overview of the FME workflow, from ingesting BD TOPO, LiDAR, and Raster data to Shapefile export

6. Results and Validation

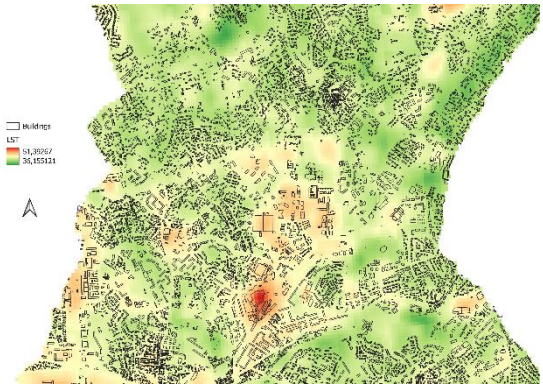


Figure 2: LST result of 23/07/2013 cut down at the level of the 13th arrondissement

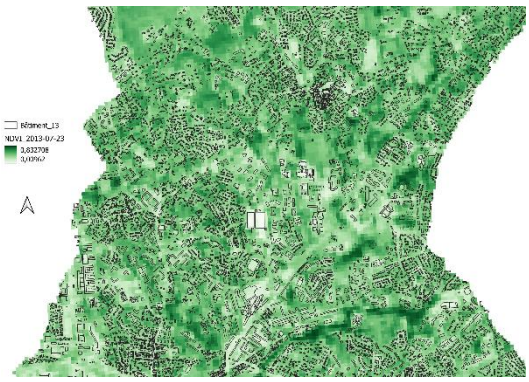
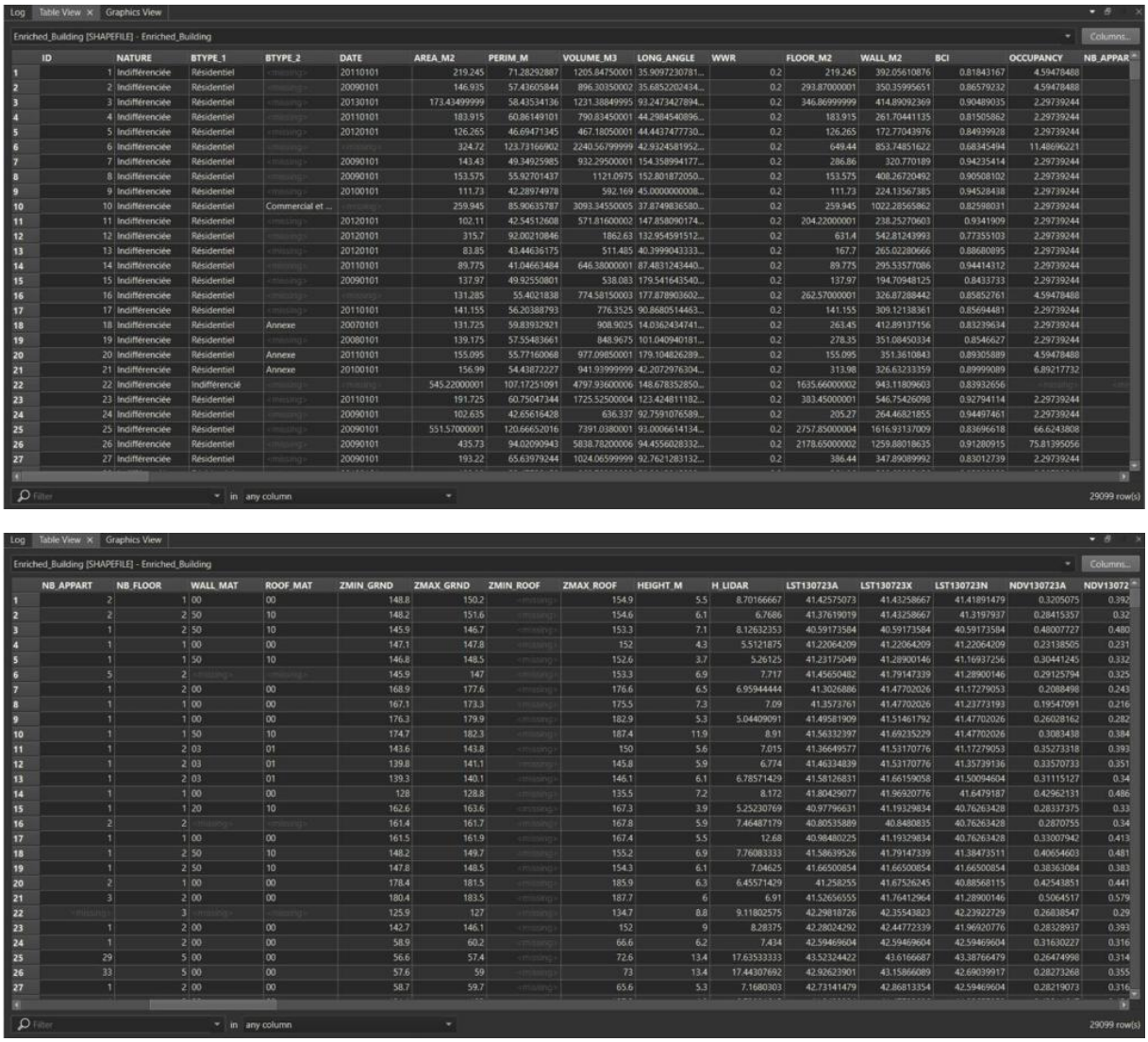


Figure 3: Result of the NDVI of 23/07/2013 cut down at the level of the 13th arrondissement



ID	NATURE	BTYPE_1	BTYPE_2	DATE	AREA_M2	PERIM_M	VOLUME_M3	LONG_ANGLE	WWR	FLOOR_M2	WALL_M2	BCI	OCCUPANCY	NB_APPAR
1	Indifférenciée	Résidentiel	<missing>	20110101	219.245	71.28292887	1205.84750001	35.9097230781...	0.2	219.245	392.05610876	0.81843167	4.59478488	
2	Indifférenciée	Résidentiel	<missing>	20090101	146.935	57.43605844	896.30350002	35.6852202434...	0.2	293.87000001	350.35995651	0.86579232	4.59478488	
3	Indifférenciée	Résidentiel	<missing>	20130101	173.43499999	58.43534136	1231.38849995	93.2473427894...	0.2	346.86999999	414.89092369	0.90489035	2.29739244	
4	Indifférenciée	Résidentiel	<missing>	20110101	183.915	60.86149101	790.83450001	44.2984540896...	0.2	183.915	261.70441135	0.81505862	2.29739244	
5	Indifférenciée	Résidentiel	<missing>	20120101	126.265	46.69471345	467.18050001	44.4437477730...	0.2	126.265	172.77043976	0.84939928	2.29739244	
6	Indifférenciée	Résidentiel	<missing>	<missing>	324.72	123.73166902	2240.56799999	42.9324581952...	0.2	649.44	853.74851622	0.68345494	11.48696221	
7	Indifférenciée	Résidentiel	<missing>	20090101	143.43	49.34925985	932.29500001	154.358994177...	0.2	286.86	320.770189	0.94235414	2.29739244	
8	Indifférenciée	Résidentiel	<missing>	20090101	153.575	55.92701437	1121.0975	152.801872050...	0.2	153.575	408.26720492	0.90508102	2.29739244	
9	Indifférenciée	Résidentiel	<missing>	20100101	111.73	42.28974978	592.169	45.0000000008...	0.2	111.73	224.13567385	0.94528438	2.29739244	
10	Indifférenciée	Résidentiel	Commercial et ...	<missing>	259.945	85.90635787	3093.34550005	37.8749836580...	0.2	259.945	1022.28565862	0.82598031	2.29739244	
11	Indifférenciée	Résidentiel	<missing>	20120101	102.11	42.54512608	571.81600002	147.858090174...	0.2	204.22000001	238.25270603	0.9341909	2.29739244	
12	Indifférenciée	Résidentiel	<missing>	20120101	315.7	92.00210846	1862.63	132.954591512...	0.2	631.4	542.81243993	0.77355103	2.29739244	
13	Indifférenciée	Résidentiel	<missing>	20120101	83.85	43.44636175	511.485	40.3999043333...	0.2	167.7	265.02280666	0.88680895	2.29739244	
14	Indifférenciée	Résidentiel	<missing>	20110101	89.775	41.04663484	646.38000001	87.4831243440...	0.2	89.775	295.53577086	0.94414312	2.29739244	
15	Indifférenciée	Résidentiel	<missing>	20090101	137.97	49.92550801	538.083	179.541643540...	0.2	137.97	194.70948125	0.8433733	2.29739244	
16	Indifférenciée	Résidentiel	<missing>	<missing>	131.285	55.4021838	774.58150003	177.878903602...	0.2	262.57000001	326.87288442	0.85852761	4.59478488	
17	Indifférenciée	Résidentiel	<missing>	20110101	141.155	56.20388793	776.3525	90.8680514463...	0.2	141.155	309.12138361	0.85694481	2.29739244	
18	Indifférenciée	Résidentiel	Annexe	20070101	131.725	59.83932921	908.9025	14.0362434741...	0.2	263.45	412.89137156	0.83239634	2.29739244	
19	Indifférenciée	Résidentiel	<missing>	20080101	139.175	57.55483661	848.9675	101.040940181...	0.2	278.35	351.08450334	0.8546627	2.29739244	
20	Indifférenciée	Résidentiel	Annexe	20110101	155.095	55.77160068	977.09850001	179.104826289...	0.2	155.095	351.3610843	0.89305889	4.59478488	
21	Indifférenciée	Résidentiel	Annexe	20100101	156.99	54.43872227	941.93999999	42.2072976304...	0.2	313.98	326.63233359	0.89999089	6.89217732	
22	Indifférenciée	Indifférencié	<missing>	<missing>	545.22000001	107.17251091	4797.93600006	148.678352850...	0.2	1635.66000002	943.11809603	0.83932656	<missing>	<miss
23	Indifférenciée	Résidentiel	<missing>	20110101	191.725	60.75047344	1725.52500004	123.424811182...	0.2	383.45000001	546.75426098	0.92794114	2.29739244	
24	Indifférenciée	Résidentiel	<missing>	20090101	102.635	42.65616428	636.337	92.7591076589...	0.2	205.27	264.46821855	0.94497461	2.29739244	
25	Indifférenciée	Résidentiel	<missing>	20090101	551.57000001	120.66652016	7391.0380001	93.0006614134...	0.2	2757.85000004	1616.93137009	0.83696618	66.6243808	
26	Indifférenciée	Résidentiel	<missing>	20090101	435.73	94.02090943	5838.78200006	94.4556028332...	0.2	2178.65000002	1259.88018635	0.91280915	75.81395056	
27	Indifférenciée	Résidentiel	<missing>	20090101	193.22	65.63979244	1024.06599999	92.7621283132...	0.2	386.44	347.89089992	0.83012739	2.29739244	

NB_APPART	NB_FLOOR	WALL_MAT	ROOF_MAT	ZMIN_GRND	ZMAX_GRND	ZMIN_ROOF	ZMAX_ROOF	HEIGHT_M	H_LIDAR	LST130723A	LST130723X	LST130723N	NDV130723A	NDV13072
2	1	00	00	148.8	150.2	<missing>	154.9	5.5	8.70166667	41.42575073	41.43258667	41.41891479	0.3205075	0.392
2	2	50	10	148.2	151.6	<missing>	154.6	6.1	6.7686	41.37619019	41.43258667	41.3197937	0.28415357	0.32
1	2	50	10	145.9	146.7	<missing>	153.3	7.1	8.12632353	40.59173584	40.59173584	40.59173584	0.48007727	0.480
1	1	00	00	147.1	147.8	<missing>	152	4.3	5.5121875	41.22064209	41.22064209	41.22064209	0.23138505	0.231
1	1	50	10	146.8	148.5	<missing>	152.6	3.7	5.26125	41.23175049	41.28900146	41.16937256	0.30441245	0.332
5	2	<missing>	<missing>	145.9	147	<missing>	153.3	6.9	7.717	41.45650482	41.79147339	41.28900146	0.29125794	0.325
1	2	00	00	168.9	177.6	<missing>	176.6	6.5	6.95944444	41.3026886	41.47702026	41.17279053	0.2088498	0.243
1	1	00	00	167.1	173.3	<missing>	175.5	7.3	7.09	41.3573761	41.47702026	41.23773193	0.19547091	0.216
1	1	00	00	176.3	179.9	<missing>	182.9	5.3	5.04409091	41.49581909	41.51461792	41.47702026	0.26028162	0.282
1	1	50	10	174.7	182.3	<missing>	187.4	11.9	8.91	41.56332397	41.69235229	41.47702026	0.3083438	0.384
1	2	03	01	143.6	143.8	<missing>	150	5.6	7.015	41.36649577	41.53170776	41.17279053	0.35273318	0.393
1	2	03	01	139.8	141.1	<missing>	145.8	5.9	6.774	41.46334839	41.53170776	41.35739136	0.33570733	0.351
1	2	03	01	139.3	140.1	<missing>	146.1	6.1	6.78571429	41.58126831	41.66159058	41.50094604	0.31115127	0.34
1	1	00	00	128	128.8	<missing>	135.5	7.2	8.172	41.80429077	41.96920776	41.6479187	0.42962131	0.486
1	1	20	10	162.6	163.6	<missing>	167.3	3.9	5.25230769	40.97796631	41.19329834	40.76263428	0.28337375	0.33
2	2	<missing>	<missing>	161.4	161.7	<missing>	167.8	5.9	7.46487179	40.80535889	40.8480835	40.76263428	0.2870755	0.34
1	1	00	00	161.5	161.9	<missing>	167.4	5.5	12.68	40.98480225	41.19329834	40.76263428	0.33007942	0.413
1	2	50	10	148.2	149.7	<missing>	155.2	6.9	7.76083333	41.58639526	41.79147339	41.38473511	0.40654603	0.481
1	2	50	10	147.8	148.5	<missing>	154.3	6.1	7.04625	41.66500854	41.66500854	41.66500854	0.38363084	0.383
2	1	00	00	178.4	181.5	<missing>	185.9	6.3	6.45571429	41.258255	41.67526245	40.88568115	0.42543851	0.441
3	2	00	00	180.4	183.5	<missing>	187.7	6	6.91	41.52656555	41.76412964	41.28900146	0.5064517	0.579
<missing>	3	<missing>	<missing>	125.9	127	<missing>	134.7	8.8	9.11802575	42.29818726	42.35543823	42.23922729	0.26838547	0.29
1	2	00	00	142.7	146.1	<missing>	152	9	8.28375	42.28024292	42.44772339	41.96920776	0.28328937	0.393
1	2	00	00	58.9	60.2	<missing>	66.6	6.2	7.434	42.59469604	42.59469604	42.59469604	0.31630227	0.316
29	5	00	00	56.6	57.4	<missing>	72.6	13.4	17.63533333	43.52324422	43.6166687	43.38766479	0.26474998	0.314
33	5	00	00	57.6	59	<missing>	73	13.4	17.44307692	42.92623901	43.15866089	42.69039917	0.28273268	0.355
1	2	00	00	58.7	59.7	<missing>	65.6	5.3	7.1680303	42.73141479	42.86833354	42.59469604	0.28219073	0.316

Figure 2: Extract from the attribute table of the Enriched_Building.shp file illustrating the spectral, altimetry and usage fields

Point	Observation	Treatment
Geometries	29,099 polygons, none invalid, empty or out of bounds	Controlled by scripts/validate_submission.py
Spectral Statistics	9,600 values (100 buildings × 32 bands × 3 statistics) recalculated outside FME	Concordance with FME; min order ≤ average ≤ max verified
Materials and date	WALL_MAT and ROOF_MAT NULL for 60% of buildings, DATE for 17,326	No reliable source: NULL kept and documented

Point	Observation	Treatment
Occupancy	Σ = 108,278 people compared to 91,365 (INSEE), i.e. +18.5%	Coefficient of 127.1 m^2 per dwelling to be recalibrated
Heights	3,828 LiDAR / BD TOPO deviations > 5 m; 151 BCIs > 1; 72 null volumes (H = 0)	Flagged for review prior to simulation
Landsat Resolution	30 m pixels for 41 m^2 median buildings	Neighbourhood values, no roof
Pretreatments	Cutting BD TOPO / OSM to the 13th floor and production of DSM / DTM made in separate FME workspaces	workflow.fmw reads this prepared data

The generated file compiles all 120 required fields (including geometries, 4 dates × 8 indices × 3 statistics, as well as typologies) while guaranteeing a healthy topological format and without unjustified null values. The combination of FME and Python scripts makes it possible to fully meet the urban modeling specifications.